

Bio-Dimension™
Software Manual

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Section 1

Introduction and Installation

1.1 About This Manual

This manual describes the installation and operation of the Bio-Dimension computer software.

The manual is organized in the following manner:

Section 1 introduces the software and describes the installation procedures.

Section 2 is a tutorial that familiarizes the user with the Detector Control Program.

Section 3 is a tutorial covering the Data Reduction Program and advanced analytical techniques.

1.2 Introduction to the Software

The Bio-Dimension computer software combines the advanced technology of the Bio-Dimension detector with the power of an IBM AT or compatible computer to produce the state of the art in high speed scanning and multiple wavelength detection systems.

The Bio-Dimension software requires Microsoft Windows 286 or 386, version 2.1 or greater, to run. Because it uses a "windows" format, the software is easy to learn and use. Additionally, you may run it simultaneously with any other software program designed to run under Microsoft Windows.

The complete software package consists of four files: DET-RUN.EXE, DETECTOR.EXE, RDC.EXE, and PARABEN.BFF. DETECTOR.EXE controls DET-RUN.EXE, together these files run the first of the Bio-Dimension programs, the Detector Control Program. RDC.EXE runs the second program, the Data Reduction Program. PARABEN.BFF contains sample data for use with the tutorial.

The capabilities of the Detector Control Program and the Data Reduction Program are introduced below.

The Detector Control Program is used for routine operation. The program increases the inherent power in the detector by supplying the following capabilities:

- Remote control of detector functions through a computer keyboard or mouse.
- High Speed Polychrome operation which allows full spectral data, within the 190 to 800 nm range, to be monitored at a data rate of 96 points/second/wavelength.
- File storage and retrieval of detector programming parameters. In addition to annotating a particular chromatographic run, this feature allows the user to easily down load specific detector parameters for a given application.
- Automated or Batch Mode operation in which predefined detector parameter files are linked to be down loaded and performed sequentially under the control of an autosampler or other external triggering device.
- Real-time computer display of elapsed time and chromatographic data (chromatogram trace and current absorbance values) for up to three of the monitored wavelengths.

The Data Reduction Program provides the following features:

- File storage and retrieval of chromatographic (single and multiple wavelength) and spectral data for future display and reduction.
- Display and hard copy production of chromatograms and spectra. Single or multiple wavelength

chromatograms may be displayed using a two-dimensional X-Y format (absorbance vs. time) in which all wavelengths are overlaid, or a three-dimensional X-Y-Z array (absorbance vs. time vs. wavelength).

- Time Slice operation which permits the generation of time-correct spectra from multiple wavelength information by slicing the three-dimensional chromatographic data of absorbance vs. time vs. wavelength perpendicularly to the time axis. The resultant spectral resolution is dependent upon the distance between monitored wavelengths. This feature enables the examination of a peak's spectra at any point during its elution, a valuable tool in determining peak purity.
- File subtraction for point-to-point baseline correction and deconvolution of overlapping peaks.
- Wavelength absorbance ratioing for peak purity determination.
- First and second derivatives of chromatograms or spectra.
- Spectra normalization and ratioing for peak purity determination.

1.3 Software Installation

What You Will Need

To install the software you will need the following:

1. Bio-Dimension UV/Vis detector.
2. IBM AT or compatible computer. Minimum requirements are 640K RAM, parallel printer port, serial RS-232 port, hard disk, and graphic display. We recommend a computer with a clock speed at least 12 MHz.
3. IBM PC-DOS or MS-DOS (version 3.1 or greater).
4. Microsoft Windows (286 or 386, version 2.1 or greater).

The following required items are supplied when you purchase the software.

1. An envelope containing the computer software diskette.
2. Software coding device.
3. 25 pin RS-232 cable.
4. This manual.
5. Software registration form.*

Unpacking the Computer Software

In the box containing the Computer Software you should find:

1. *The Bio-Dimension Software Owner's Manual.*
2. An envelope containing the Bio-Dimension software diskette.
3. A software coding device.
4. Software product registration form.

Ensure that all items are present. If damage to any of these items is noted, immediately contact the shipping company. If any of these items prove to be missing, contact your sales representative or Bio-Rad labs. Before opening the Bio-Dimension software diskette envelope, carefully read the software license agreement located on the front of the envelope. By opening the envelope and breaking the seal, the user indicates full acceptance of all license agreement terms.

Be aware that you should always handle the diskette with care, ensuring against exposure to heat, direct sunlight, or excessive dust. Never touch the magnetic surface of the diskette. When not in use, always store the diskette, in its protective pouch, in a cool, dry area. The use of a diskette file is recommended.

Note: The Sentinel-Pro Block Coding Device is sensitive to static electricity. To prevent damage to the device, handle only the plastic parts unless you are appropriately grounded. Attach the coding device ONLY to the printer parallel port.

* The software registration form should be filled out and returned to Bio-Rad immediately to ensure that future software updates are delivered promptly.

1.4 Hardware Considerations

Computer Requirements

The software programs require the use of a computer using an 80286 microprocessor (IBM AT or compatible clone) or 80386 microprocessor, with clock frequencies of at least 8 MHz. A clock speed of 12 MHz is recommended for 80286 machines. The computer requires 640K RAM and a hard disk drive. You will also need a graphics monitor that is compatible with Microsoft Windows, such as a Hercules, EGA, or VGA graphics monitor.

RS-232 Connections

The detector connects to the computer via a RS-232 serial port. Detector connections require a 25 pin, male RS-232 plug. Typically, a computer's RS-232 port is a 9 or 25 pin male jack. If your computer has a 25 pin male RS-232 jack, interfacing should be accomplished by using a RS-232 straight extension cable provided with the software. If your computer uses a 9 pin male jack, interfacing requires the use of a RS-232 25 to 9 pin cable. These cables are readily available at most computer sales outlets.

Identify the RS-232 port on the rear of the detector. Using the correct cable, connect the male end of the RS-232 cable to the detector RS-232 port. Be sure to tighten the fastening screws. Connect the 9 or 25 pin female end of the RS-232 cable to the computer RS-232 port. Be sure to tighten the fastening screws. The detector-computer interface is now complete.

Computer Mouse Connections

The use of a mouse with Microsoft Windows and the computer software is required for Bio-Dimension software. Typically, a mouse connects to the computer using one of the serial ports (a serial mouse) or through the installation of a circuit board in one of the computer's expansion slots (a bus mouse). If you desire to connect more than one detector to your computer, it is best to use a bus mouse, which leaves both serial ports available.

1.5 Coding Device Connections

The coding device protects against the use of multiple copies of the computer software while allowing the user to run without a key diskette. In order for the Detector Control Program to function, the coding device must be installed on the computer's printer parallel port. **ATTACH ONLY TO THE PARALLEL PORT.** Identify the printer parallel port on your computer. Typically, it is a female 25 pin subminiature "D" connector.

Install the coding device on the parallel port. One end of the coding device is labeled "computer," ensure this end connects to the port. Fasten the tightening screws. Connect the printer input cable to the output of the "coding device." Be sure to tighten the fastening screws. The coding device will NOT affect the operation of the printer.

1.6 Software Considerations

DOS

If you have not already done so, install IBM PC-DOS or Microsoft MS-DOS on the hard disk drive of your computer. Check with your *DOS Manual* for more information on the installation and setup of DOS on your computer.

Microsoft Windows

If you have not already done so, install Microsoft Windows (286 or 386) versions 2.1 or greater. This item is not provided with the Bio-Dimension package. Microsoft Windows is readily available at most computer software sales outlets. Familiarize yourself with the procedures for operating under Windows. Windows

operating procedures will not be addressed in this manual. For answers to questions regarding Microsoft Windows, consult the *Microsoft Windows Owner's Manual*.

Making a Back-Up/Working Diskette

In order to protect the integrity of the original Bio-Dimension software diskette, it is recommended that a back-up or working copy of the programs be made. It should be noted that multiple copies of the software can be made, however they will not function without the installation of the "coding device."

Turn on the computer. After the DOS prompt is displayed, insert the computer software diskette into drive A and place a blank, formatted diskette into drive B. Enter the command: copy A:*. * B:*. *. If you have only one disk drive, it will be necessary to switch diskettes back and forth until all files are copied. Verify that all files were copied to drive B. Enter the command: dir b:. There should be four files copied: DETECTOR.EXE, DET-RUN.EXE, RDC.EXE, and PARABEN.BFF. For more information on copying and backing up files, consult your *DOS Manual*.

Installing the Software

For organizational sake, the files should be copied into a directory dedicated to the detector. You may also copy them into the same subdirectory as the Bio-Rad HRLC® software.

Copy DETECTOR.EXE, DET-RUN.EXE, RDC.EXE, and PARABEN.BFF into the Bio-Dimension software directory by entering the following commands: Copy A:*. * D:\HRLC*. *. Verify that all files were copied by checking the contents of the directory. Remove your diskettes and store in a safe place.

Activating the Software

In order to activate the Bio-Dimension software, you must first activate the Microsoft Windows program. The Windows directory should be located in the path statement of your AUTOEXEC.BAT file. If it is not, refer to the *Microsoft Windows Manual*. If you are currently running series 500 and 800 software, Windows is already correctly installed.

Change to the Bio-Dimension software directory. Activate the Windows program by entering the command: win. The display now shows all the files located under the Bio-Dimension directory.

After turning on the detector, activate the DETECTOR.EXE file by double-clicking on it with the mouse. The display will now show the Detector Setup main menu.

Detector Setup: (Untitled)

Files Start

Detector Mode

> High Speed Scanning <

Multi Wavelength

Single Wavelength

Descriptions

Display

Options

Sequence

Number of Runs: 1

Output File: paraben

MEMO

DATE:	October 18, 1994	CC:	David Greenwood
TO:	Zen Torres, Mark Smith, Destry Miller		
FROM:	Sylvia L. Arias-Montez		
SUBJECT:	Alternate Bio-Dimension software installation		

Dear Colleagues:

Insert this into page 4 of the software manual as an alternate installation.

The Bio-Dimension software installation instructions are on page 4 of the software manual, and they stipulate placement of the Bio-dimension software to be installed in the HRLC sub-directory. In order to do create an icon on the desktop, I recommend the following (The actual commands are in **boldface**):

1. Using DOS, type the following after the C prompt: **md bio-dime**. Hit return. This will make a sub-directory called bio-dime. Verify that the directory has been made by typing C:\ **dir/w** or **dir /p**. The /w means that the directory will be displaced in a wide format, the /p for page format. These are useful commands when you have a lot of information on the C drive and a lot of info whizzes by on the screen.
2. Install the Bio-Dimension diskette into the A drive (or whatever you call this drive).
3. Change from the C:\ drive to the A:\ drive by typing C:\a:\ and hit return.
4. Type **copy *.* C:\bio-dime**, hit return. This will copy the 4 files from the Bio-Dimension diskette into the bio-dime sub-directory.
5. Next, go into Program Manager (Windows) by returning to your C drive (type a:\c:), hit return. At the C:\ type **win** to take you into Windows.
6. Within the Program Manager window, select "New" from the File Menu. Select New Program Group. Type **Bio-Dimension** in the Description box. The group file box can remain empty. This will create a window (program group) called Bio-Dimension.
7. While the Bio-dimension window is open, select "New" from the File Menu and New Program Item.

In the Description box, type **Bio-Dimension**.

In the Command Line box, type **C:\bio-dime\detector.exe**.

The file, detector.exe is the executable file to start the bio-dimension software.

In the Working Directory box, type **C:\bio-dime**.

In the Shortcut Key, type **None**.

Select Change Icon button on the right-hand side of the window, to select your icon. Click on OK. This will create an icon within the program group. By double-clicking on this icon, you can start the program.

The RDC.EXE program is activated in the same manner.

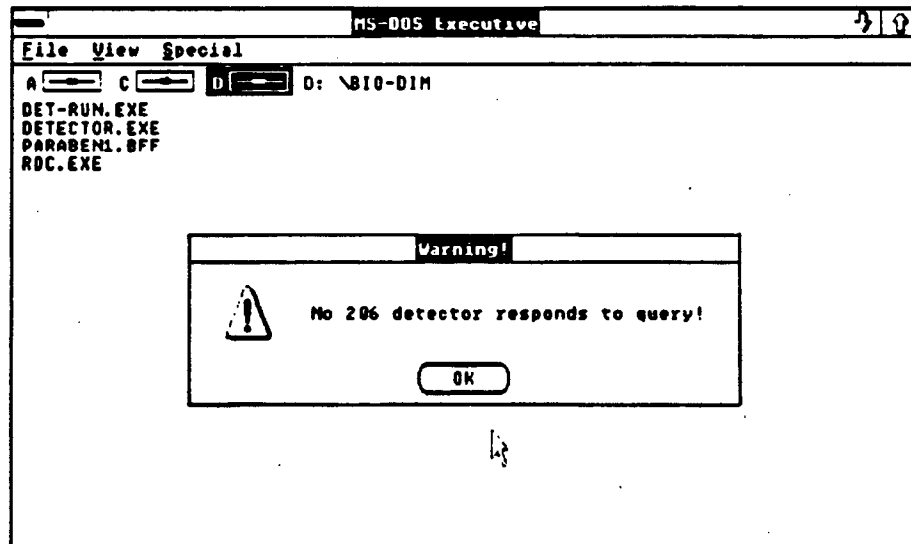
You now know how to activate the Detector Control and Data Reduction Programs. Each program can run separately, or they can run simultaneously. To run the programs at the same time, activate both in the usual manner. Then resize and set the active window as discussed in the *Microsoft Windows Manual*.

Note: On 8 and 10 MHz 80286 based computers, you may notice reduction in software running speed while running both DETECTOR.EXE and RDC.EXE.

Possible Error Messages

There are two possible warnings that may be displayed upon activating the Detector Control Program.

If you fail to turn on the detector before activating DETECTOR.EXE, or if the RS-232 cable is not correctly installed, the following error message will be displayed:



If you should activate the Detector Control Program without the coding device installed on the computer's printer parallel port, the following error message will appear:

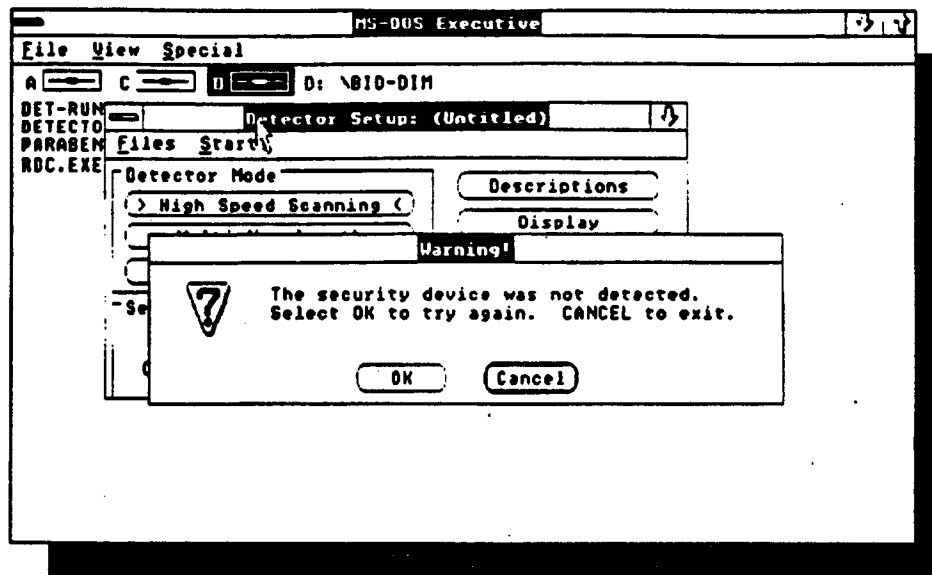


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Section 2

The Detector Control Program Tutorial

The Bio-Dimension requires Microsoft Windows to run, the user is instructed to refer to the *Microsoft Windows Owner's Manual* for details regarding basic operating procedures. Such procedures are not addressed in this manual.

This manual only covers the operation of the detector software package. The user should already be familiar with the operational characteristics of the detector before using the software. The required information is covered in the *Bio-Dimension Detector Owner's Manual*.

For simplicity's sake, this manual will consider the operation of the Detector Control and Data Reduction Programs one at a time. However, it is possible to operate the programs simultaneously. See the "Activating the Software" portion of the preceding section for further details.

2.1 The Detector Control Program

Software Overview

- Perform High Sensitivity Monochrome
- Perform Multi -Wavelength
- Perform High Speed Scanning
- Schedule Monochrome Wavelength changes to occur during the run
- Perform a Full Spectral Autozero at any time during the run
- Initiate an Event Mark on recorder outputs
- Alter the status of a number of auxiliary items
- Set automatic detector lamp shut off
- Manually control the deuterium lamp status
- Manually control the tungsten lamp status
- Activate the detector's Zero Upon Wavelength Change function

Post processing is done with the Data Reduction Program. The Data Reduction Program will be discussed in Section 3.

HELPFUL HINTS...

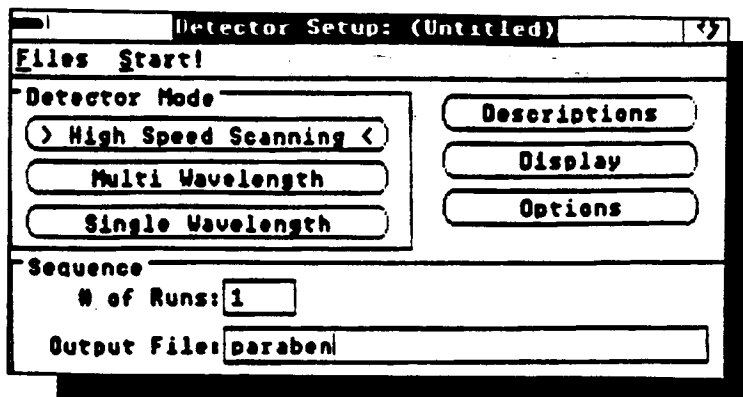
For the initial stage of methods development, in which the sample is being injected for the first time, it is best to scan all possible wavelengths extending from just above the solvent's cut off point to the maximum expected significant wavelength value. This is best accomplished using the High Speed Polychrome function. This allows the user to see all peaks, along with their maximum wavelengths of absorbance, present in the sample.

After optimizing solvent conditions for maximum peak resolution, the user may continue to utilize High Speed Polychrome or, if only a few wavelengths corresponding to absorbance maxima are required, change to High Sensitivity Polychrome or Monochrome.

2.2 Detector Setup Main Menu

To start the program, change to the directory with the DETECTOR.EXE file in it and type "win Detector." Windows must be in the DOS "path" statement to launch DETECTOR EXE. If you encounter errors refer to your *DOS Manual* under the "path" command.

When the program is first started you see the DETECTOR SETUP main menu. From this menu all of the detector's capabilities can be realized. For the purpose of this tutorial we will first perform High Speed Scanning. Then we will perform a Single Wavelength and a Multiple Wavelength run. This tutorial will demonstrate all of the main features of the Bio-Dimension and give helpful operating hints. For the purposes of this tutorial it is not necessary to actually perform chromatography.



Before collecting data, you should ensure the detector is configured for the correct COM port.

1. Select OPTIONS from the main menu.
2. Set COM port to correct port number.
3. Click OK.

Failure to configure the COM port will result in an error. See Section 1.6 "Possible Error Messages".

2.3 High Speed Scanning

Instructions at a Glance:

1. Program spectral parameters using the High Speed Scanning menu.
2. Choose the wavelengths to display using the display menu.
3. Name the output file—from the bottom of the main menu.
4. Click START! on the main menu line.

You may want to fill in the description section. Any description entered will stay with that particular chromatogram. It is not important to enter a description for this tutorial, and the description information can be changed at a later date.

Each of these steps is discussed in detail in the following paragraphs.

Detailed Instructions:

STEP 1: CLICK THE HIGH SPEED SCANNING BOX

We are going to perform a chromatogram in which a wide range of the spectrum may be simultaneously monitored.

High Speed Scanning Menu

Each scan is defined by specifying start, stop, and step size wavelength values. The start wavelength is lowest wavelength to be monitored in the scan. The stop wavelength is the highest wavelength to be monitored in the scan. The step size value is the interval existing between scanned wavelengths. As an example, a High Speed Scanning program using 200 nm, 360 nm, and 5 nm as the start, stop, and step size parameters will monitor 200, 205, 210, 215 . . . 360 nm.

Note: The entered start and stop values must be within their allowed wavelength limits by a factor at least as large as the step size divided by two. For example, a step size of 6 requires a UV scan to have a start wavelength of 193 ($190 + 6/2$) and a stop wavelength of 362 ($365 - 6/2$). If you attempt to enter 190 and 365 as the start and stop wavelengths using a 6 nm step size, an error message will appear.

Note: The spectral range of the instrument extends from 190 to 800 nm. However, scans crossing the deuterium and tungsten lamp cross-over are not permitted. Thus, scans must be in the UV/Vis (190-365 nm) or Visible (366-800 nm) regions.

If you are performing chromatography, select a start wavelength value which exceeds the UV-cutoff of your eluant. If you are not running a column, use the following parameters for start, stop, and step size wavelengths, respectively: 220, 320, 5 nm.

Move the cursor to the START field and enter the start wavelength (220 nm).

Move the cursor to the STOP field and enter the stop wavelength (320 nm).

Move the cursor to the STEP SIZE field and enter the step size wavelength value (5 nm).

Advance to the STOP TIME field and enter a time occurring shortly after the elution of the last peak. If you are not performing chromatography, the default value of 10 minutes will suffice.

If you have used the suggested parameters, the HIGH SPEED SCANNING window should look like this:

Detector Setup: (Untitled)

Files Start!

Detector M High Speed Scanning

High Sp

Multi

Single

Sequence

of R

Output F

Start: 220 nm

Stop: 320 nm

Step Size: 5 nm

Stop Time: 10.0 mins

Options

Play

ons

OK Cancel

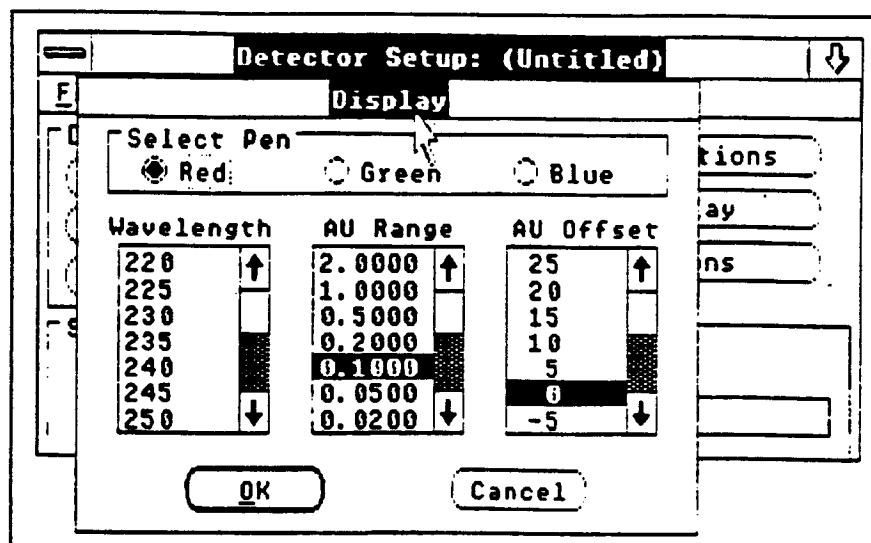
When you are satisfied with your selections, activate the OK button.

STEP 2: CHOOSE THE WAVELENGTHS TO DISPLAY

Display Menu

After the spectra is defined you must set up the display menu. You can display up to three wavelengths at a time. During a run, wavelengths and absorbance range can be changed.

Click on the DISPLAY box.



You will note the specified wavelengths present in the WAVELENGTH listbox.

1. Using the keyboard or the mouse, select an AU RANGE appropriate for your test mix.

REMEMBER: The absorbance range can be easily changed during a run, so if you are not sure of the range select 0.002 AUFS for starters.

2. Select an AU OFFSET percentage.

The screen real-time display is capable of displaying three different traces. You may display only one, or you can look at the same wavelength at three different ranges. To select further traces:

Activate the GREEN pen button and select the desired wavelength and range for the green pen. If you are not performing chromatography, enter 0.005 AUFS. Activate the BLUE pen button and select the desired wavelength and range for the blue pen. If you are not performing chromatography, enter 0.001 AUFS.

When you are satisfied with your selections, activate the OK button.

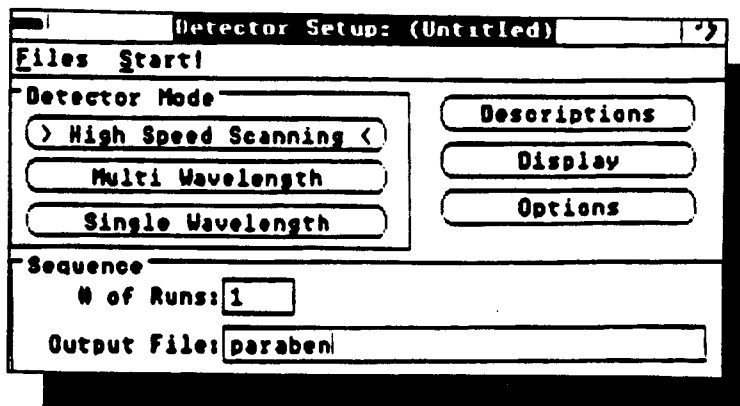
Note: The set up pens routine only controls parameters for CRT real-time display. It does not affect the detector recorder outputs or the raw data stored on the hard disk.

STEP 3: NAME THE OUTPUT FILE

You must now enter the name of the file in which the data will be stored. Enter the name in the box labeled "OUTPUT FILE" on the bottom of the main menu. For each run, the Bio-Dimension software will add a number to the file name for easy recall of stored data. For example if your output file was called "paraben" the first three runs would be called:

PARAB001.BFF
PARAB002.BFF
PARAB003.BFF

The file name will be truncated to be in accordance with MS-DOS requirements, and the DOS extension "BFF" will be added.



STEP 4: CLICK START! When you are ready to begin, simply select START! from the main menu line. The COLLECTION display will open and you will be monitoring the baseline.

2.4 The Collection Display

The collection display has a small box with absorbances superimposed on a strip chart. The absorbance box is a Window's "child window" and can be moved like any window or removed by double clicking the control menu box.

While collecting data, the collection display control box provides the ability to change the pens being displayed, to start and stop a run, and to autozero the detector.

The collection will monitor the baseline until it receives an event mark or begin collection signal.

After the event signal, the detector will autozero and data will be collected for the time specified in the HIGH SPEED SCANNING menu. The data will be stored in the file you selected in the main menu.

During a run you may change absorbance ranges, change wavelengths, and stop a run. Simply select the option from the window control box. You may also hide the absorbance box from the window control box. For Multiple Wavelength and Monochrome operation, you may place event marks on the recorder outputs using the EVENT command in the window control box.

After the run is finished you may view it with the Data Reduction Program (RDC.EXE). See Section 3 for details.

2.5 Saving a Program File

Once you set up High Speed Scanning, simply select SAVE AS. . . in the control menu box. Enter a program name when prompted and all of your detector parameters will be saved. When you call up DETECTOR.EXE again, you can open this file from the control menu box and all your operating parameters will be in effect, thus eliminating the need to reprogram everyday. The file name you specify will be stored with the DOS extension "PGM."

2.6 Single Wavelength Mode

Single Wavelength operation is essentially the same as High Speed Scanning. The only difference is in setting up the wavelength to be monitored.

Instructions at a Glance:

The steps for Single Wavelength are:

1. Program spectral parameters using the Single Wavelength dialog box.
2. Choose the wavelengths to display using the display dialog box.
3. Name the output file—from the bottom of the main menu.
4. Click START! on the main menu line.

The single wavelength process is explained in detail below.

Detailed Instructions:

STEP 1: SELECT SINGLE WAVELENGTH FROM THE MAIN MENU

Single Wavelength - Line Programmable

Wavelength: 254 ☒ Zero upon change

From: 0.0 To: 10.0

Chart Recorders

Channel: 1

AU Range: 0.5000

Rise Time: 1.0

Time	Wavelen	Chan	Range
10.0	254	1	0.5000, 1.0

If you are performing chromatography, enter a wavelength appropriate for the solutes in your test mix. To do so, move the cursor to the WAVELENGTH (nm) box, delete the displayed value, and enter the value of choice.

Move the cursor to the stop time box (To: __) and enter the end time of your run. If you are not performing chromatography, leave the default value of 10 minutes.

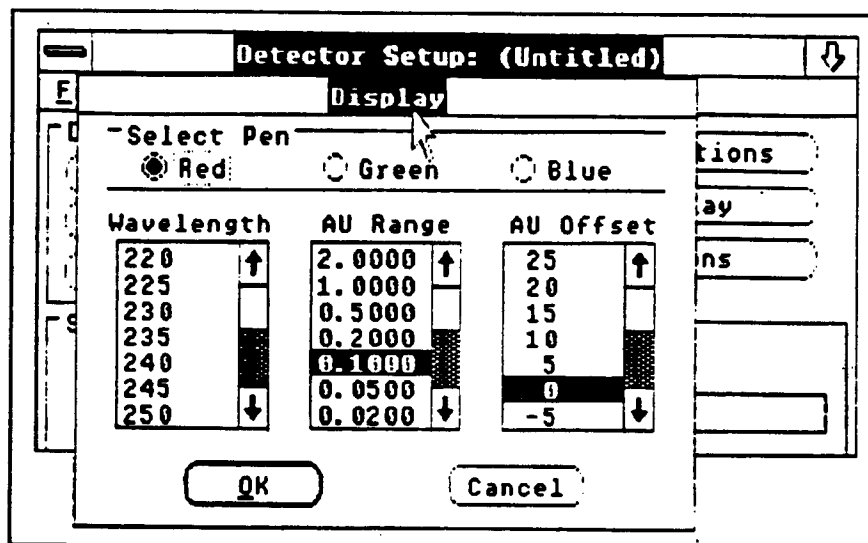
The Bio-Dimension software allows for control of the recorder outputs on the rear of the instrument. If you would like to control the recorder outputs perform the following steps:

1. Move the cursor to the CHANNEL box and enter the desired recorder output channel on the detector for the transmission of real-time data.
2. Move the cursor to the AU RANGE box and enter a range appropriate for your test mix. The range will be used as AUFS on the recorder output channel.
3. Move the cursor to the RISE TIME box and enter a rise time appropriate for your test mix. For information regarding proper rise time selection, consult the *Bio-Dimension Detector Owner's Manual*.

Once you have entered the desired wavelength click OK.

STEP 2: CHOOSE THE WAVELENGTHS TO DISPLAY

Display Menu



After the wavelength is selected you must tell the display routine to display it. As with High Speed Scanning, wavelengths and absorbance range can be changed during a run.

Click on the DISPLAY box.

You will note the specified wavelength present in the WAVELENGTH listbox.

1. Using the keyboard or the mouse, select the desired wavelength.
2. Select a range appropriate for your test mix.

REMEMBER: The absorbance range can be easily changed during a run, so if you are not sure of the range select 0.002 AUFS for starters.

3. Select an AU OFFSET percentage.

The screen real-time display is capable of displaying three different traces. You may display only one, or you can look at the same wavelength at three different ranges. For the first pen select an AU range from the AU RANGE box. You may also select an AU OFFSET. To select further traces:

Activate the GREEN pen button and select the desired wavelength and range for the green pen. If you are not performing chromatography, enter 0.005 AUFS. Activate the BLUE pen button and select the desired wavelength and range for the blue pen. If you are not performing chromatography, enter 0.001 AUFS.

When you are satisfied with your selections, activate the OK button.

Note: The set up pens routine only controls parameters for CRT real-time display. It does not affect the detector recorder outputs or the raw data stored on the hard disk.

STEP 3: NAME THE OUTPUT FILE

You must now enter the name of the file in which the data will be stored. Enter the name in the box labeled "OUTPUT FILE" on the bottom of the main menu. For each run, the Bio-Dimension software will add a number to the file name for easy recall of stored data. For example if your output file was called "paraben" the first three runs would be called:

PARAB001.BFF

PARAB002.BFF

PARAB003.BFF

The file name will be truncated to be in accordance with MS-DOS, and the DOS extension "BFF" is added to the file name.

STEP 4: CLICK START! When you are ready to begin, simply select START! from the main menu line. The COLLECTION display will open and you will be monitoring the baseline.

The collection display is covered in this tutorial in the Section 2.4 of this manual.

2.7 Performing a Multiple Wavelength Chromatogram

We are going to perform a Multiple Wavelength Chromatogram in which a maximum of 32 different wavelengths may be monitored.

It is not necessary to perform chromatography to proceed with this section of the tutorial. You may simply store the baseline signal for now.

The steps for Multiple Wavelength operation are essentially the same as the steps for Single Wavelength operation. The major differences being selecting additional wavelengths and assigning them to recorder outputs.

Instructions at a Glance:

Multiple Wavelength operation requires the following steps:

1. Program spectral parameters using the Multiple Wavelength dialog box.
2. Choose the wavelengths to display using the display menu.
3. Name the output file—from the bottom of the main menu.
4. Click START! on the main menu line.

Notice that steps 2, 3, and 4 are exactly the same as in High Speed Scanning and Monochrome operation. Multiple Wavelength operation is explained in detail below.

Detailed Instructions:

STEP 1: SELECT THE MULTIPLE WAVELENGTH DIALOG BOX

The Multiple Wavelength dialog box appears when you click on the MULTIPLE WAVELENGTH box. This dialog box is identical to the Single Wavelength dialog box.

If you are performing chromatography, you should enter the desired wavelengths for your test mix, substituting their values for the ones presented in this tutorial. If you are not performing chromatography, use 254 and 280 nm.

The wavelength insertion box changes the highlighted wavelength to the wavelength specified. For example, enter 214 in the wavelength box and click insert: the highlighted wavelength is now changed to 214 nm. The insert command also inserts a new, additional wavelength (215), 1 nm greater than the value you entered. Now the new wavelength (215) is highlighted and you may enter parameters for it. Enter 254 in the wavelength box, and the new highlighted wavelength will be set to 254. If you choose, you may select up to 32 wavelengths to monitor.

Note: The scanning rate for Multiple Wavelength operation is 24 points/second/wavelength.

Advance the cursor to the STOP TIME (minutes) field. Enter a time occurring shortly after the elution of the last peak. If you are not performing chromatography, the default value of 10 minutes should suffice.

If you have entered all parameters as outlined in this tutorial, the MULTIPLE WAVELENGTH dialog box should look like this:

Detector Setup: (Untitled)

Files Start!

Detector Mode: Multi Wavelength

Wavelength: 254 [Insert] [Remove]

Chart Recorders

Channel: 1

AU Range: 0.0200

Rise Time: 1.0

Wavelength	Chan	Range	RT
254	1	0.0200	1.0
280	2	0.0200	1.0

Ratio

Channel: 4 254 nm over 280 nm

Integrator Output Wavelength: 280

Stop Time: 10.0

[OK] [Cancel]

You may optionally direct any of the absorbances out the recorder outputs. To do so:

1. Highlight the desired wavelength and, in the AU RANGE box, enter the full scale range appropriate for your test mix at this wavelength.
2. Assign a recorder channel in the CHANNEL box.
3. Move the cursor to the RISE TIME field and enter an appropriate rise time.

The newly entered wavelength along with its channel output, range, and rise time will be displayed in the WAVELENGTH listbox.

You may delete a wavelength in the WAVELENGTH listbox. Deletion is accomplished by activating the line and selecting the REMOVE button.

Ratio Plots

If you choose, you may plot ratios on one of the recorder outputs. To do so, advance the cursor to the CHANNEL field of the RATIO listbox. Enter the desired detector recorder output channel for the ratio trace. Channel four is suggested. Advance the cursor to the NM box. Enter the wavelength desired to be the numerator in the ratio trace. Advance to the next NM box and enter the wavelength chosen to be the denominator in the ratio trace. If you are using the suggested wavelengths, ratio 254 over 280.

Integrator Output

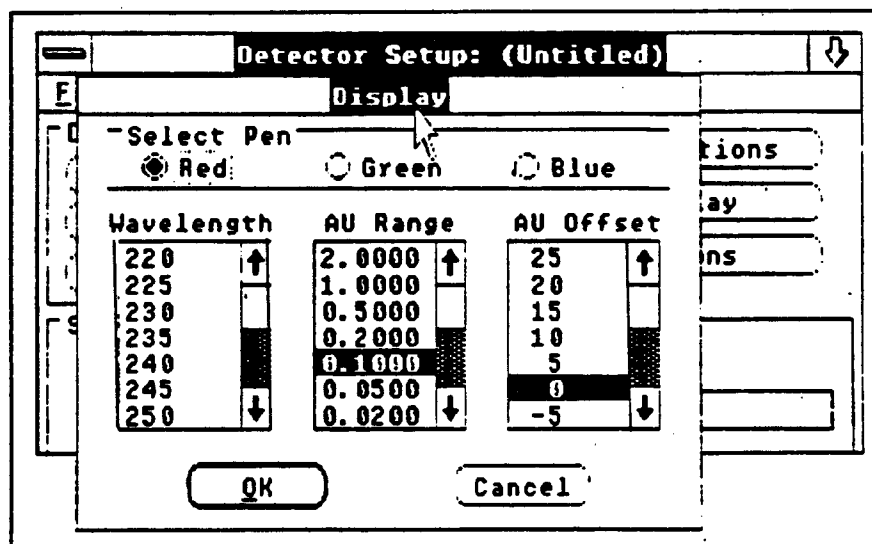
Finally, move the cursor down to the INTEGRATOR OUTPUT WAVELENGTH (nm) field and enter the desired wavelength to be transmitted out of the detector's real-time integrator output: 254 nm is suggested.

Note: The integrator output is a fixed sensitivity output of 1 V/AU. Event marks will not appear on the integrator output.

When you are satisfied with your selections, activate the OK button.

STEP 2: CHOOSE THE WAVELENGTHS TO DISPLAY

Display Menu



After the wavelengths are selected you must tell the display routine which wavelengths to display. As in High Speed Scanning, wavelengths and absorbance range can be changed during a run.

Click on the DISPLAY box.

You will note the specified wavelengths in the WAVELENGTH listbox.

1. Using the mouse, select the desired wavelength.
2. Select a range appropriate for your test mix.

REMEMBER The absorbance range can be easily changed during a run, so if you are not sure of the range select 0.002 AUFS for starters.

3. Select a display offset percentage.

The screen real-time display is capable of displaying three different traces. You may display only one, or you can look at the same wavelength at three different ranges. For the first pen, select the desired range under the AU RANGE box. To select further traces:

Activate the GREEN pen button and select the desired wavelength and range for the green pen. If you are not performing chromatography, enter 0.005 AUFS. Activate the BLUE pen button and select the desired wavelength and range for the blue pen. If you are not performing chromatography, enter 0.001 AUFS.

When you are satisfied with your selections, activate the OK button.

Note: The set up pens routine only controls parameters for CRT real-time display. It does not affect the detector recorder outputs or the raw data stored on the hard disk.

STEP 3: NAME THE OUTPUT FILE

You must now enter the name of the file in which to store the data. Enter the name in the box labeled "OUTPUT FILE" on the bottom of the main menu. For each run, the Bio-Dimension software will add a number to the file name for easy recall of stored data. For example if your output file was called "paraben" the first three runs would be called:

PARAB001.BFF
PARAB002.BFF
PARAB003.BFF

The file name will be truncated to be in accordance with MS-DOS, and the DOS extension "BFF" will be added.

STEP 4: CLICK START! When you are ready to begin, simply select START! from the main menu line. The COLLECTION display will open and you will be monitoring the baseline.

Operation of the collection display is covered in the Section 2.4.

2.8 Additional Software Features

There are a few additional capabilities not yet covered in the tutorial; they are described in this section:

Options Menu

This dialog box allows for remote control of a number of detector parameters. All items are selected by activating the box, using either the computer keyboard or mouse.

The SCHEDULED ZEROS text box allows for scheduled absorbance zeros. Enter each time, in minutes, when you would like the chromatogram to be zeroed. Separate each scheduled time with a comma. Remember the chromatogram is automatically zeroed at the beginning of each run.

The TURN LAMPS OFF . . . check box is used to shut off the detector's lamps at the end of a run or sequence of runs.

Detector Setup: (Untitled)

Options

Scheduled Zeros:

Conn Port (1,2):

Rise Time: seconds

☐ Turn Lamps off at end of all runs.

OK Cancel

Batch Operation

The Batch Mode of operation allows you to automatically collect multiple chromatographic runs. Each injection may be automatically triggered through the detector's remote event closure. Consult the *Detector Manual* for more information on setting up the external event.

Under SEQUENCE in the main menu is a box labeled "# of runs." Simply enter the desired number of times you will perform the chromatogram.

Detector Setup: (Untitled)

Files Start!

Detector Mode

> High Speed Scanning <

Multi Wavelength

Single Wavelength

Descriptions

Display

Options

Sequence

of Runs:

Output Files:

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Section 3

The Data Reduction Program Tutorial

3.1 The Data Reduction Program

The Data Reduction Program is used to visualize, manipulate, and print chromatographic and/or spectral data collected through the use of the Detector Control Program. The Data Reduction Program provides the following features:

- File storage and retrieval of chromatographic (single and multiple wavelength) and spectral data for future display and reduction.
- Display and hard copy production of chromatograms and spectra. Single or multiple wavelength chromatograms may be displayed using a two-dimensional X-Y format (absorbance vs. time, in which multiple wavelength signals are overlaid), or a three-dimensional X-Y-Z array (absorbance vs. time vs. wavelength).
- Time Slice operation which permits the formation of time-correct spectra from multiple wavelength information by slicing the three-dimensional chromatographic data of absorbance vs. time vs. wavelength perpendicularly to the time axis. This feature enables the examination of a peak's spectra at any point during its elution.
- File subtraction for baseline correction and deconvolution of overlapping peaks.
- Wavelength absorbance ratioing for peak purity determination.
- First and second derivatives of chromatograms or spectra.
- Spectra normalization and ratioing for peak purity determination.

It is not necessary to have the detector connected to the computer nor to have the software coding device attached to run the Data Reduction Program.

The Data Reduction Program is the file labeled RDC.EXE. To bring up the software, simply activate the RDC.EXE file in the same way DETECTOR.EXE was activated.

3.2 Displaying a Chromatogram

Instructions at a Glance:

1. Open the file using the FILES pull down menu.
2. Select the display option of the CHROMATOGRAM pull down menu.
3. Choose the files to display and the display range.
4. Click OK.

Detailed instructions for displaying a Chromatogram follow:

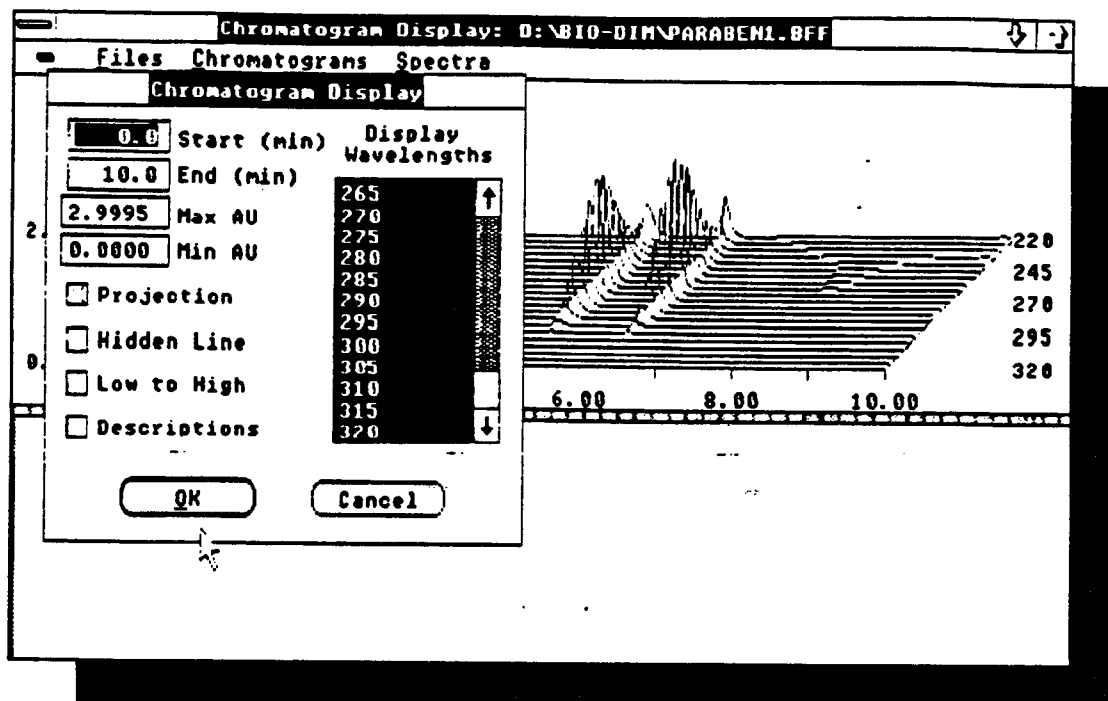
Detailed Instructions:

STEP 1: ACTIVATE THE FILES PULL DOWN MENU AND SELECT THE OPEN OPTION

The SELECT FILE dialog box should appear. It will list all data files that you have stored. Next to each file's name is the first description field that you used to annotate the file. For this exercise select PARABEN.BFF. You may select any file in the listbox or type in the file name in the edit box. Then activate the OPEN button.

STEP 2: SELECT THE DISPLAY OPTION OF THE CHROMATOGRAMS PULL DOWN MENU

The CHROMATOGRAM DISPLAY dialog box will appear.



Move the cursor to the START (min) field and enter the start time for the displayed chromatogram. You may keep this parameter at 0 for now.

The END (min) field lists the end time for the recorded chromatogram. For now, do not change this parameter.

Move the cursor to the MAX AU field and enter an appropriate maximum range for this chromatogram. You may want to refer back to the conditions used when collecting the chromatogram. For PARABEN.BFF use 2.9995 AU.

Move the cursor to the MIN AU field and enter a minimum range, normally 0 or slightly below 0.

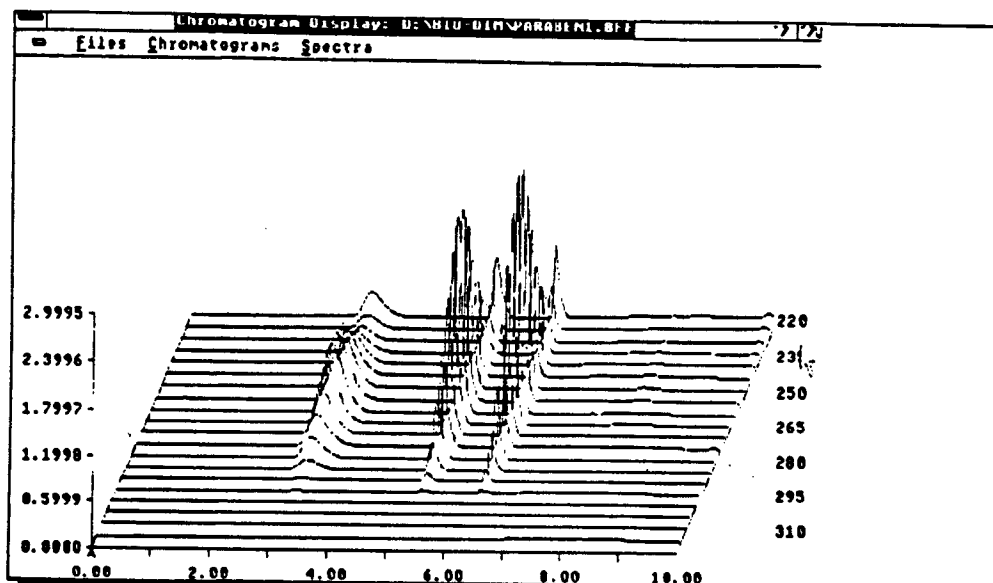
STEP 3: HIGHLIGHT THE WAVELENGTHS IN THE DISPLAY WAVELENGTHS LISTBOX THAT YOU WOULD LIKE TO VIEW

Note: Multiple wavelengths may be selected by holding down the shift key and selecting each wavelength with the mouse.

The PROJECTION, HIDDEN LINE, LOW TO HIGH, and DESCRIPTIONS check boxes allow you to modify the way the display is created. For more information, see the "Optional Display Settings" section of this manual. For this example, select the PROJECTION box.

STEP 4: ACTIVATE THE OK BUTTON

The chromatogram will now be displayed. The chromatogram is displayed in a small window within the main window. This smaller window, referred to as a child window, may be moved or maximized within the main window.



So far, we have displayed the full chromatographic trace; however, it is possible to display only a portion of the chromatogram. Of course this can be done using different maximum and minimum ranges and start and end times. In the example (PARABEN.BFF), a trace of a 10 minute chromatogram is displayed. Note the group of closely eluting peaks at 4 to 7 minutes. For a better view, zoom into the area at 4 to 7 minutes.

To do so, one would activate the CHROMATOGRAPH DISPLAY dialog box, enter start and stop times of 4 and 7 minutes respectively, and then activate the OK field. The new chromatographic trace will be displayed.

HELPFUL HINTS....

Often, the display of too many wavelength traces make the chromatogram difficult to read. In such cases, it is prudent to redisplay the chromatogram using a reduced number of wavelengths. To do so, activate the CHROMATOGRAPH DISPLAY dialog box. While holding down the shift key, use the mouse to delete some of the displayed wavelengths.

Note: While a large number of wavelengths is displayed a large amount of computer memory is dedicated to the display of that window. This can reduce the amount of memory available to other tasks. Once the number of displayed wavelengths is reduced or the child window is closed the memory is made available to other tasks.

3.3 Optional Display Settings

Activate the DISPLAY option of the CHROMATOGRAMS pull down menu. Try changing the check boxes and viewing the resulting effects. Select OK to view after each change. PROJECTION, HIDDEN LINE, and LOW TO HIGH only work when multiple wavelengths are displayed.

Descriptions

The option displays the chromatogram's descriptions. These descriptions were made during the run, however you may alter them using RDC.EXE software. For convenience, the first line of the description field is displayed when you select "open file" in the RDC.EXE software.

Projection

This option displays the chromatogram wavelengths in a pseudo three-dimensional graph. The graph gives you a combination of the spectral and chromatographic information.

Hidden Line

The HIDDEN LINE option removes lines of the chromatogram that would be obscured by peaks. Selection of HIDDEN LINE slows display of the chromatogram, but it can give a less cluttered display in the areas around peaks.

Low to High

The option reverses the wavelength axis in the display. Because most compounds exhibit greater absorbance at shorter wavelengths, the default setting for the wavelength axis is from longer to shorter wavelengths. This is done to keep large peaks from obscuring smaller ones. But, sometimes, it is best to display the chromatogram with the wavelength axis running from shorter to longer wavelengths. To do so, set the LOW TO HIGH check box.

Explore different ways to view the chromatograms. Once the data is collected, RDC.EXE will not permanently alter the data.

3.4 Making a Printed Copy of the Chromatogram

Activate the pull down menu under the rectangular bar located at the top left corner of the child window. (You must have the chromatogram displayed on the screen.) If the child window is maximized within the main window use the bar to the left of the FILES menu item.

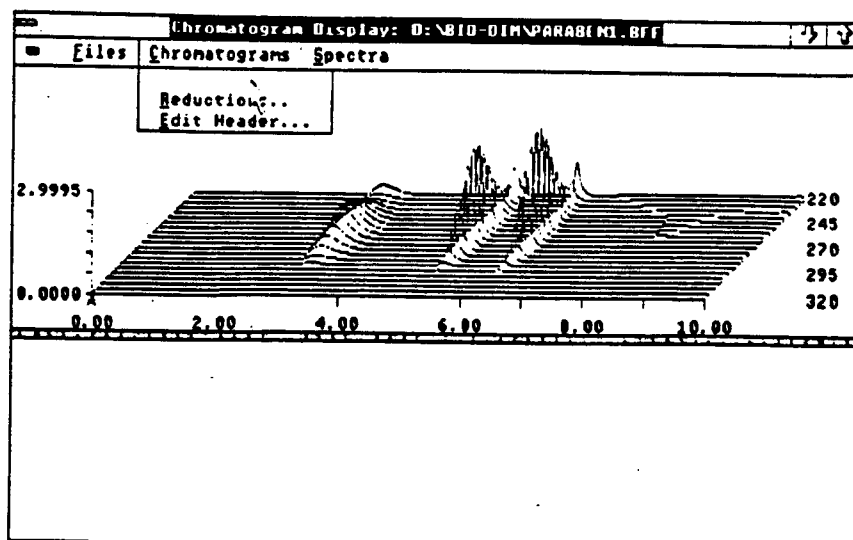
Select the PRINT option. A printer dialog box will be displayed allowing you to set your print options. The printer dialog box is dependent on the printer that you have set up for Windows. A second box will be displayed allowing you to cancel the print. After a brief pause the print is routed to the Windows print spooler. The printer will begin printing a hard copy of the displayed chromatogram.

The Reduction Program prints chromatograms using the same aspect ratio as on the computer screen. To change the printed chromatogram, resize the window and select PRINT again.

3.5 Analyzing Chromatograms

We are going to look at the analysis of chromatograms. If RDC.EXE is not loaded, load it in the usual manner. Open the chromatogram file PARABEN.BFF as outlined in Section 3.2 of this manual.

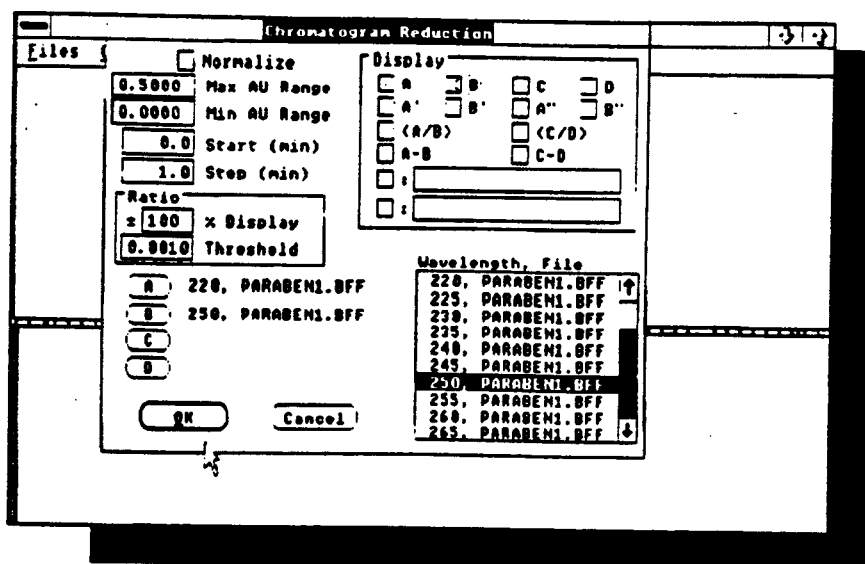
Select the REDUCTION option of the CHROMATOGRAMS pull down menu. The CHROMATOGRAM REDUCTION dialog box will be displayed.



The CHROMATOGRAM REDUCTION dialog box allows you to select up to four wavelengths (A, B, C, and D) from two files. These chromatograms may be displayed with DISPLAY options on the right side of the dialog box. The wavelengths' listbox is filled with the available wavelengths from two files, the main file, selected by the OPEN option of FILES pull down menu, and the auxiliary file, selected by the AUXILIARY option of the FILES pull down menu. For information on opening auxiliary files see Section 3.11.

First select the wavelength:

Highlight the wavelength in the listbox and then press the WAVELENGTH letter button. The wavelength and the file name will be displayed to the right of the lettered button. Select each wavelength you would like to analyze. After selecting a wavelength, you must choose one of the display options.



Display Options

The first four options A, B, C, and D will display the respective chromatogram's wavelength. A' and B' will display the 1st derivatives of the chromatograms A and B. A'' and B'' will display the 2nd derivatives of the chromatograms A and B. Derivatives of C and D are displayed in a similar manner.

Note: All the displayed derivatives are normalized to the same scale and then displayed. This will cause A' to change size when displayed if A' is added or removed from the display options.

Options <A/B> and <C/D> display ratios of A over B and C over D. The ratio graphs are normalized so that each segment (normally each peak) is centered vertically on the screen. If the brackets are removed, the segments at the peaks are not centered.

Options A-B and C-D are the differences of two chromatograms.

The last two options are user definable equations. User input formulas may combine several of the above options. The user equations must be written with parentheses indicating operator precedence.

Examples of user defined equations might be:

(A-B)'	derivative of a difference
(A-B)/(C-D)	ratio of two differences
C''	2nd derivative of C

The allowed operators are:

+	addition
-	subtraction
*	multiplication
/	division
'	first derivative
''	second derivative
</>	centered ratio

Another major display option is the **NORMALIZE** check box. When you check this option all **DISPLAY** options that you check are normalized to full screen before being displayed. This feature is useful when comparing the shapes of two different wavelengths' derivatives. Below on the left are the non-normalized 1st derivatives. On the right are the same derivatives, normalized.

MAX AU RANGE and **MIN AU RANGE** set display ranges for the chromatograms and chromatogram differences. **START (MIN)** and **STOP (MIN)** set the time limits over which the chromatograms will be read and displayed.

Ratio Parameters

On the left of the dialog box are two ratio parameters. The **Ratio THRESHOLD** sets the minimum AU value at which both chromatograms must be set before ratioing is started. Ratioing is stopped when either chromatogram falls below 75% of the threshold. The threshold prevents the ratioing of the noise on the baseline between the chromatogram peaks. Potentially large distortions of the ratio occur when either value is close to zero.

The second ratio parameter is the **%DISPLAY**. This value sets the display range for ratios. The range is set at the average ratio value plus and minus the average ratio value times the **%DISPLAY**. So if you enter a **%DISPLAY** of 50 and the average value of the ratio graph is 1.2 (indicating that a peak is 1.2 times taller at the numerator wavelength than at the denominator wavelength) then the ratio will be displayed on a scale of 0.6 to 1.8. This is $1.2 \pm (1.2 * 0.50)$.

After you have selected the desired options select the **OK** button.

Display Colors

The graphs are displayed in four colors; black for the chromatograms and differences; red for derivatives; blue for centered ratios; and green for ratios.

3.6 Creating and Displaying a Time Slice Spectra

We are going to select, display, and save a Time Slice spectra. The **Time Slice** function generates time-correct spectra by slicing **Multiple Wavelength** chromatograms perpendicular to the time axis. The **Time Slice** function uses a two-dimensional cubic spline fit to determine spectra.

The spectral resolution of **Time Slice** spectra is dependent upon the wavelength interval between monitored wavelengths for the selected **Multiple Wavelength** file. If the **Multiple Wavelength** data has more than five monitored wavelengths and was collected in **High Speed Polychrome**, the **RDC.EXE** software is able to spline the data in the wavelength direction creating smooth spectra.

A **Time Slice** is a valuable tool in the determination of peak purity. With this option, the user can generate spectra at any time during the elution of a given peak. These spectra can then be compared. If differences are noted, the peak can be considered to be impure.

Load the **Data Reduction Program** in the usual manner. Open a **High Speed** file. **PARABEN.BFF** contains **High Speed** data. **High Sensitivity Multiple Wavelength** data may also be **Time Sliced**, however there are normally only two or three wavelengths being monitored in **High Sensitivity Polychrome**.

Note: *Time Slices of less than five wavelengths are generally not as useful as Time Slices with more wavelengths.*

Instructions at a Glance:

To display a Time Slice Spectra:

1. Display the chromatogram using the method outlined in Section 3.1.
2. Select Time Slice display from the SPECTRA pull down menu.
3. Move the mouse to the zero time on the chromatograph, press and hold the mouse button and drag the Time Slice line to the desired time.
4. To save a Time Slice for later analysis, select Time Slice List.

Display of Time Slice spectra is described in detail below.

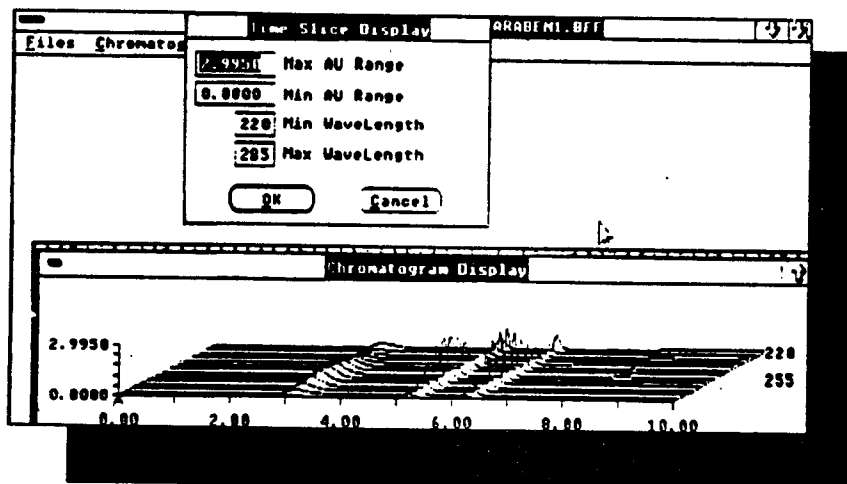
Detailed Instructions:

STEP 1: DISPLAY THE CHROMATOGRAM USING THE METHOD OUTLINED IN SECTION 3.1 OF THIS MANUAL

The chromatogram start and end display times should be set to display the area of interest. If the peaks are very narrow on the screen then you might want to set the start and end display times to get a better view of the peaks.

STEP 2: SELECT TIME SLICE DISPLAY FROM THE SPECTRA PULL DOWN MENU

The TIME SLICE DISPLAY dialog box will be displayed.



The default settings are from the display parameters. If you choose you may alter these settings. Move the cursor to the MAX AU RANGE field and enter an appropriate range for the Time Slice. Move the cursor to the MIN AU RANGE field and enter an appropriate range for the Time Slice.

Move the cursor down to the MIN WAVELENGTH field and enter the minimum wavelength value for the Time Slice.

Move the cursor to the MAX WAVELENGTH field and enter the maximum wavelength for the Time Slice.

STEP 3: ACTIVATE THE OK FIELD

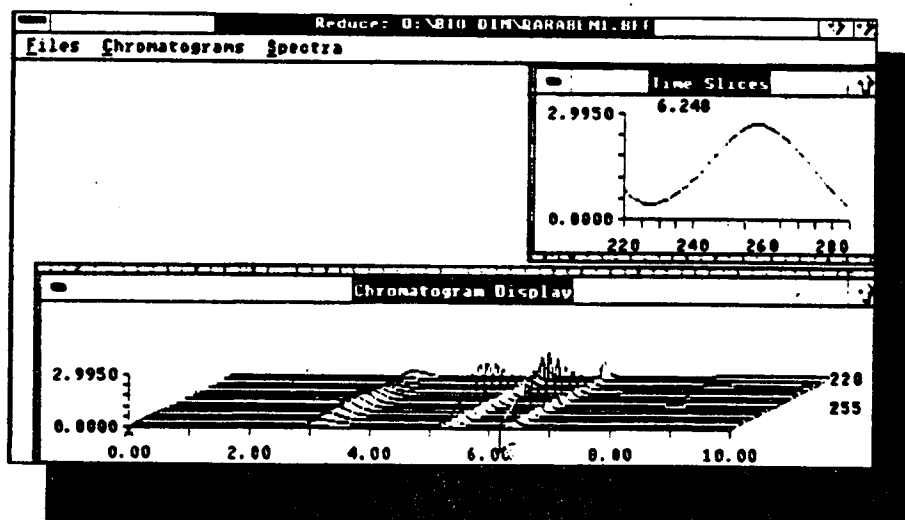
The Time Slice will be displayed in an additional child window. The default location of the child window is the upper right side of the main window. You may move the child window or change its size using the mouse.

STEP 4: MAKE THE CHROMATOGRAM DISPLAY WINDOW ACTIVE

Click the mouse within the child window. Time Slices are created by pressing the mouse button and dragging the red line to the position where you would like the Time Slice. When you first activate the Time Slice display, the red line is at time zero.

STEP 5: MOVE THE MOUSE TO TIME ZERO AND PRESS AND HOLD THE BUTTON

Move the mouse to the right. Notice the slice time is continually updated as you move the mouse. When you release the mouse button, the Time Slice at the current position is read from the data file and is displayed in the Time Slice window. You may again press the mouse button and move the Time Slice to another position.



3.7 Saving the Time Slice for Later Analysis

Saving a Time Slice will allow reduction operations to be performed on spectra. Derivative plots, spectral subtraction, and other operations can be performed. The "Analyzing Spectra" section of this manual has more information. Time Slices must be added to the Time Slice list in order to perform analysis.

When you have displayed the desired slice you may save it to the chromatogram file. Select the ADD TIME SLICE option from the SPECTRA pull down menu. You will be asked if you would like to add the Time Slice at a certain time to the file. Select OK to save the slice.

HELPFUL HINTS....

An additional method to add the Time Slice is to select the TIME SLICE LIST option from the SPECTRA pull down menu. The TIME SLICE LIST dialog box will be displayed containing a listbox of the current file's Time Slices. It also indicates the current slice. You may add the Time Slice by pressing the ADD TO LIST button. You may also remove existing Time Slices from the file by selecting the Time Slice in the listbox and pressing the REMOVE FROM LIST button. When you are done adding and/or removing Time Slices from the file press the EXIT button.

Time Slices that are added to the Time Slice list may be overlaid and further analyzed using the Spectra Reduction feature. This will be covered in the "Analyzing Spectra" section of this manual.

3.8 Analyzing Spectra

We are going to now look at the analysis of spectra. If the Data Reduction Program (RDC.EXE) is not loaded, load it in the usual manner.

Open a chromatogram file. Select the REDUCTION option of the SPECTRA pull down menu. The SPECTRA REDUCTION dialog box will be displayed along with the available Time Slices. If no Time Slices are available see Section 3.6, "Creating and displaying a Time Slice Spectra." Only Spectra from the Time Slice list can be analyzed.

Spectra Reduction

Normalize

Max AU Range: 0.5000
Min AU Range: 0.0000
Min Waven: 190
Max Waven: 800

Display

☐ A ☐ B ☐ C ☐ D
☐ A' ☐ B' ☐ C' ☐ D'
☐ (A/B) ☐ C/D
☐ A-B ☐ C-D
☐ :
☐ :

Ratio

x Display: 100
Threshold: 0.0010

A 5.226 220-320 Slice PARABEN1.BFF
B
C
D

Minutes	nm	Type	File
3.044	220-320	Slice	PARABEN1.BFF
5.226	220-320	Slice	PARABEN1.BFF
6.248	220-320	Slice	PARABEN1.BFF
5.245	220-320	Slice	PARABEN1.BFF

OK
Cancel

The SPECTRA REDUCTION dialog box allows you to select up to four spectra (A, B, C, and D) from two files. These spectra may be displayed with DISPLAY options, on the right side of the dialog box; display options are similar to the chromatogram reduction options. The spectra listbox is filled with the available Time Slices from two files, the main file, selected by the OPEN option of the FILES pull down menu, and, if selected, the auxiliary file, selected by the AUXILIARY option of the FILES pull down menu. For this exercise, only Time Slices from the main file will be used. For information on opening auxiliary files see Section 3.11.

Select the spectrum. Highlight a spectrum in the listbox and press the SPECTRA LETTER button. The spectra time; start, stop, and step wavelengths; and the file name will be displayed to the right of the letter button. Select each spectrum you would like to analyze.

Select the Display Parameters. The first four options A, B, C, and D will display the respective spectra. A' and B' will display the 1st derivatives of the spectra A and B. A'' and B'' will display the 2nd derivatives of the spectra A and B. Derivatives of C and D are displayed in a similar manner.

Note: All the displayed derivatives are normalized to the same scale and then displayed. This will cause A' to change size when displayed if A' is added or removed from the display options.

Options A-B and C-D are the differences of two spectra.

The last two options are user definable equations. User input formulas may combine several of the above options. The user equations must be written with parentheses indicating operator precedence.

Examples of user defined equations might be:

$(A')-(B')$	difference of two derivatives
$(A-B)/(C-D)$	ratio of two differences
C''	2nd derivative of C

The allowed operators are:

+	addition
-	subtraction
*	multiplication
/	division
'	first derivative
"	second derivative
</>	ratio with similarity index

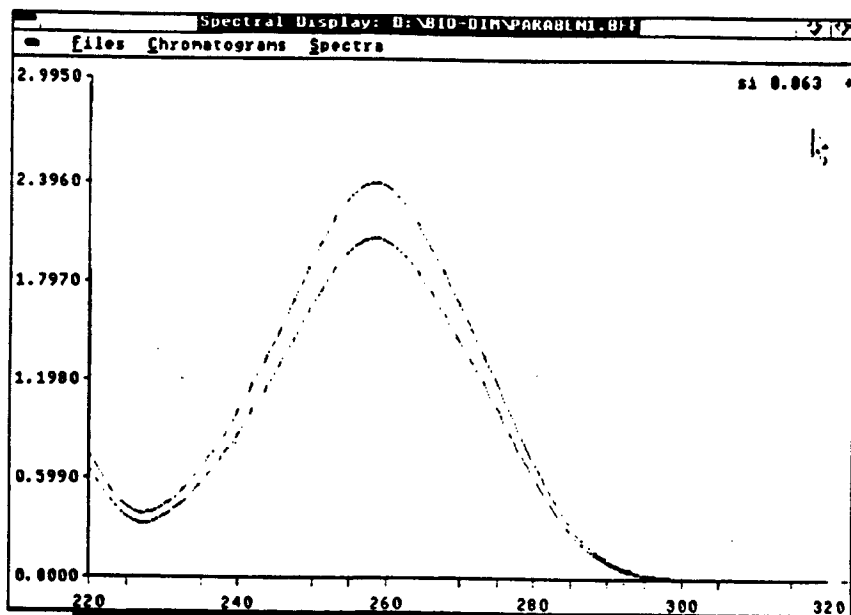
Another major display option is the NORMALIZE check box. When you check this option all DISPLAY options that you checked are normalized to full screen before being displayed. This feature is useful when comparing the shapes of two different spectra's derivatives. Below on the left are the non-normalized 1st derivatives. On the right, are the same derivatives, normalized.

MAX AU RANGE and MIN AU RANGE set display ranges for the spectra and spectra differences. MIN WAVELEN and MAX WAVELEN set the wavelength display limits over which the spectra will be displayed. Proper values are determined by looking at the selected spectra's start and stop wavelengths.

Similarity Index

The option <A/B> produces a ratio and a similarity index (si). The spectra A and B are ratioed. The ratio is graphed. The maximum ratio value and the minimum ratio value are ratioed (RATIOMAX/RATIOMIN or RATIOMIN/RATIOMAX) to give a similarity index (si). The similarity index can range from zero to one and is displayed in the upper right corner of the graph. Remove the brackets, as in "C/D", and only the ratio of C over D is calculated. No similarity index is calculated.

Note: The closer the similarity index is to one, the more similar the spectra.



Graph Colors

The graphs are displayed in four colors; black for the chromatograms and differences; red for derivatives; blue for normalized spectra ratios; and green for ratios.

Ratio Parameters

On the left of the dialog box are two ratio parameters. The ratio **THRESHOLD** sets the minimum AU value at which both spectra must be set before ratioing is started. Ratioing is stopped when either spectra falls below 75% of the threshold. The threshold prevents the ratioing of two values where one value is very close to zero. Potentially large distortions of the ratio occur when either value is close to zero.

The second ratio parameter is the **%DISPLAY**. This value sets the display range for ratios. The range is set at the average ratio value plus and minus the average ratio value times the **%DISPLAY**. So if you enter a **%DISPLAY** of 50 and the average value of the ratio graph is 1.2 (indicates one spectra is 20% larger than the other spectra) then the ratio will be displayed on a scale of 0.6 to 1.8. This is $1.2 \pm (1.2 * 0.50)$.

After you have selected the desired options select the OK button.

3.9 Creating an ASCII Absorbances File

The Data Reduction Program will allow you to send the absorbance values from the chromatogram file (binary) to an ASCII file that you specify. In RDC.EXE open the chromatogram file by selecting the OPEN option from the FILES pull down menu. Once the file is opened select the TO ASCII option from the FILES pull down menu.

Move the cursor to the OUTPUT TO field. Enter the name of the file that will contain the ASCII absorbances. Press the OK button.

You should consider that the ASCII file will contain the information in an uncompressed format. The ASCII file could be several times the size of the original chromatogram file. A typical ASCII output file will have the following format:

D:\BIO-DIMPARABEN.BFF
02-01-90 11:34 AM

220 230 240 250 260 270 280 290
0.0001 0.0001 0.0001 0.0002 0.0002 0.0002 0.0001 0.0001
0.0001 0.0001 0.0002 0.0002 0.0002 0.0003 0.0002 0.0002

3.10 Printing a Brief Summary of the Chromatogram Information

The Data Reduction Program will allow you to print a quick summary of the currently opened chromatogram. In RDC.EXE open the chromatogram file by selecting the OPEN option from the FILES pull down menu. Once the file is opened select the REPORT option from the FILES pull down menu. The descriptions and wavelengths monitored are summarized in the report.

3.11 File by File Subtraction

The Data Reduction Program will allow you to subtract two chromatogram files from each other and save the result into a third chromatogram file. This feature enables you to collect a chromatogram with a gradient and, post run, remove the effects of the gradient on the baseline absorbances.

Using DETECTOR.EXE you will need to collect two chromatograms. The first chromatogram is only a background run with the desired gradient. The second chromatogram contains the injected sample. The chromatograms need to be collected with the same wavelengths and over the same gradient.

Once you have collected the two chromatograms load RDC.EXE, if it is not already loaded. Select the OPEN option from the FILES pull down menu. Choose the sample chromatogram file from the listbox. Press the OPEN button.

Select the AUXILIARY option from the FILES pull down menu. Choose the background chromatogram file from the listbox. Press the OPEN button.

To subtract the two files select the SUBTRACT option from the FILES pull down menu.

The dialog box contains the two file names. Move the cursor to the OUTPUT TO field. Enter the name of the chromatogram file that will be the result of the difference of the main and auxiliary files. Press the OK button.

The subtraction is started. The subtraction process deactivates the Data Reduction Program. You may switch to other applications while RDC.EXE is subtracting the two files.

3.12 Changing the Chromatogram Description Header

Select the EDIT HEADER option. The header that you wrote prior to the collection of this file is displayed.

If you would like to modify any entry in this header, simply type in the desired modifications and press the SAVE button. The new header is now written to disk. Press the CANCEL button, if you do not want any changes, or if you do not want to save any changes that you have made.

The first line of the description header is displayed when you select "open file" from the main menu. This is an easy way to annotate and select specific chromatograms.

